

Ivan Vashchenko

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Education

Bachelor of Science in **Mechanical Engineering and Computer Science** | Principia College | May 2027

Bachelor of Science in **Mechanical Engineering** | University of North Dakota | May 2027

GPA: 3.5

Design of Machinery | Finite Element Analysis | CFD | Software Architecture | Machine Component Design | Manufacturing Processes
Electrical Engineering Circuit | Thermodynamics | Fluid Mechanics | Heat and Mass Transfer | Aerodynamics | Data Structures

Experience

Mechanical Engineering Intern | WTI Services Pure Air | Clearwater, FL & Remote | May 2024 – Present

- Supported mechanical engineering projects through **on-site assessments**, mechanical drawings, fan selection, ductwork layout, and pressure calculations using **AutoCAD**.
- Built and deployed **engineering automation tools** that consolidated multiple internal workflows, reducing manual effort, and significantly improving turnaround time.
- Led development of a **3D modeling and visualization workflow** using photogrammetry, parametric modeling, and Three.js to reconstruct large air-handling units.
- Delivered tools that allowed engineers to **compare existing vs. restored conditions**, improving design clarity and client communication.
- Managed projects independently during the academic year, coordinating directly with engineering leadership to expand and scale internal tools.
- Expanded **3D and BIM automation tools** for engineering design, integration into **CAD/Revit workflows**.
- Coordinating with engineering leadership to scale software tools into broader project delivery.

Projects

Digitally Manufactured Reactor — Mechanical Design & Fabrication

- Designed a complete mechanical assembly using a fully **CAD-driven workflow** in SolidWorks and translated into fabrication.
- Integrated **3D-printed, laser-cut, and machined components** into a single functional assembly.
- Delivered a **physical prototype** demonstrating manufacturability-driven design decisions.

BDX Robotics Project – Applied Computational Modeling Club

- Led a student team to design and prototype a robotics system integrating **mechanical design with embedded programming**.
- Focus on real-world deployment: automation, mobility, and hardware-software synergy.

Autonomous Sorting & Packaging System — Mechanical Integration Project

- Designed and fabricated mechanical subsystems for an automated bearing sorting and packaging system
- Integrated the mechanical system with a **Raspberry Pi-based vision system**, coordinating actuation with an ML-trained classification model.

Solar Car Linkage & Structural System

- Designed **linkage mechanisms** and structural components for a **solar vehicle system**.
- Evaluated tradeoffs between geometry, load paths, manufacturability, and assembly.

Material Tension Test — Brass Specimens

- Designed and executed a mechanical tension test for brass specimens.
- Established test procedures, loading conditions, and measurement approach.
- Collected and analyzed experimental data; summarized results and limitations.

Technical Skills

CAD & Design: AutoCAD | Revit | SolidWorks | ANSYS | 3DF Zephyr | Blender | Rhino

Embedded & Computing: Raspberry Pi | Lidar | C++ | C# | Java | Python | JavaScript | Three.js

Manufacturing & Prototyping: Machining | Sheet Metal | Laser Cutting | 3D printing

Leadership

Founder and President | Applied Computational Modeling Club (ACMC) | 2025 – Present

- Student organization focused on applying computational modeling to real-world engineering problems.

House President | Principia College | 2024 – Present

- Coordinating operations, budgeting, and communication for a residential dorm community.

President | Christian Science Organization | 2025 – Present

- Organization of 100+ members, coordinating events cross-functional teams, fostering an inclusive community.

The National Society of Leadership and Success (NSLS) | 2025